Reading a paper

Márton Balázs[1]

University of Bristol

Heilbronn Doctoral Partnership Career Development Event
Bristol, 15 May, 2024.

[1] Thanks go to Tim Browning and Francesco Mezzadri for inspiration.

What to look into?

Why read it?

Is this a paper worth reading?

Ok, so how to read it?

And how to write a good paper?

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- Found in search. Google is surprisingly good sometimes.
- Need to review / referee / report on it. Hopefully it's a good one and connected to your field, otherwise don't accept the job!

The problem: it's a lot!

You'll need to be aware of dozens of papers in your field (at least. . .). And which dozens out of thousands?

Carefully reading a hard paper can be as slow as 2 pages/hr. Now do the maths.

$\rightsquigarrow$ Going through a paper line by line is most often not the way to go.

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Is this a paper worth reading?

Communicating ideas and results well is difficult and very important. It can go wrong at many places.

- Maybe there are no ideas to start with. Small additions to existing results that won't bring in much novelty. (The next two were sent to me by Tim Browning:)

What? Why? Worth it? How to read? How to write?

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AMERICAN MATHEMATICAL SOCIETY
MathSciNet
Mathematical Reviews
ISSN 2167-5163

Publications results for "MR Number=(429922)"

MR0429922 (55 #2931) Reviewed
Hsu, Kuang Yau
On decomposition of certain formal groups.
Tamkang J. Math. **6** (1975), 69–78.
14L05
Get it!

Citations
From References: 0
From Reviews: 0

It is hard to imagine in a single paper such an accumulation of garbled English, unfinished sentences, undefined notions and notations, and mathematical nonsense. The author has apparently read a large number of books and papers on the subject, if one looks at his bibliography; but it is doubtful that he has understood any of them. He speaks blithely of elements of a formal group and claims to prove (!) that two formal groups having the same Lie algebra are isomorphic. One also hears of root systems of a commutative formal group, and of the union of its Borel subgroups, which belongs to the Lie algebra, etc. What is amazing to the reviewer is that such a thing was ever printed.
Reviewed by J. Dieudonne

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MathSciNet
Mathematical Reviews
ISSN 2167-5163

Publications results for "MR Number=(150536)"

MR0150536 (27 #533) Reviewed
Hewitt, Edwin; Stromberg, Karl R.
A remark on Fourier-Stieltjes transforms.
An. Acad. Brasil. Ci. **34** 1962 175–180.
42.26 (42.52)

Get it!

Citations
From References: 5
From Reviews: 0

Well-known simple theorems.
Reviewed by C. S. Herz

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Is this a paper worth reading?

- Wrong (rare): somebody, maybe you, have found an error. (More often it seems wrong at first, but it's not at a deeper look. If absolutely sure it's wrong (ask someone else around you, too, maybe supervisor) write author very politely.)

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- Citation list misses very obvious work(s).
- The paper might not be original (plagiarism): e.g. `arXiv` checks for significant text overlap. If there is one with a work from different authors, that's suspicious.

Indications it's a good paper

Ok, so how do we know if a paper is worth reading? Look for indications:

- Who is(are) the author(s)? Check `Mathscinet` if unsure.

MR: Hairer, Martin - 664196 - Vimperator

MR: Hairer, Martin - 664

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University of Bristol

Share | Edit Author Profile



Hairer, Martin

Email:	M.Hairer@Warwick.ac.uk
Website:	http://www.hairer.org/
MR Author ID:	**664196**
Earliest Indexed Publication:	2000
Total Publications:	**79**
Total Related Publications:	**9**
Total Citations:	**1210**

⊞ Published as: Hairer, M. (23)

Publications
Related Publications
Refine Search
Co-Authors
Collaboration Distance
Mathematics Genealogy Project
Citations

Co-authors (by number of collaborations)

Allman, Michael · Apte, Amit · Bass, Richard F. · Baudoin, Fabrice · Betz, Volker · Blömker, Dirk · Bou-Rabee, Nawaf M. · Burdzy, Krzysztof · Cass, Thomas · Chen, Zhen-Qing · Cloez, Bertrand · Eckmann, Jean-Pierre · Flandoli, Franco · Friz, Peter K. · Gubinelli, Massimiliano · Hairer, Ernst · Hutzenthaler, Martin · Jentzen, Arnulf · Kelly, David T. B. · Koralov, Leonid B. · Labbé, Cyril[2] · Litterer, Christian · Maas, Jan · Majda, Andrew J. · Manson, Charles · Matetski, K. · Mattingly, Jonathan C. · Ohashi, Alberto · Masayoshi F. · Pajor-Gyulai, Zsolt · Pardoux, Étienne · Pavliotis, Grigorios A. · Piatnitski, Andrey · Pillai, Natesh S. · Romito, Marco · Ryser, Marc D. · Scheutzow,

Link: http://www.ams.org/mathscinet/search/publications.html?pg1=INDI&s1=664196 http://www.ams.org/mathscinet/search/author.html?mrauthid=664196 < [1/1] All

Indications it's a good paper

- Check the paper itself on `Mathscinet`. Journal, reviews, citations (factor in year of appearance).

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ISSN 2167-5163

Publications results for "Author=(kipnis) AND Author=(varadhan)"

MR0834478 (87i:60038) Reviewed

Kipnis, C.(F-POLY)
Département de Mathématique, École Polytechnique
91128 Palaiseau, France
; Varadhan, S. R. S.(1-NY-X)
Courant Institute of Mathematical Sciences (CIMS), New York University
New York, New York, 10012

Citations
From References: 257
From Reviews: 37

Central limit theorem for additive functionals of reversible Markov processes and applications to simple exclusions.

Comm. Math. Phys. 104 (1986), no. 1, 1–19.
60F17 (60J05 60K35)

Get it!

Let $\{X_n\}_{n\geq 0}$ be a stationary, reversible, ergodic Markov chain on some state space S, with X_0 distributed according to the invariant distribution $\pi(ds)$. Let V be a function on S such that $E\{V(X_0)\} = \int V(s)\pi(ds) = 0$, $E\{V^2(X_0)\} < \infty$ and such that $\sigma^2 = \sum_{-\infty}^{+\infty} E\{V(X_0)V(X_n)\}$ converges. The authors prove that, under these weak conditions, $Y_n(t) := n^{-1/2} \sum_1^{[nt]} V(X_n)$ converges weakly to a Brownian motion $\sigma B(t)$ with variance σ^2. Note that this is proved for the stationary chain. It is more complicated if one conditions on $X_0 = x$, so that the chain is governed by the corresponding measure P_x. Under P_x, $Y_n(\cdot)$ is close to $\sigma B(\cdot)$ for x in a set S_n, with $\pi(S_n) \to 1$ as $n \to \infty$. The theorem is reduced to a central limit theorem for martingale difference sequences. Vector-valued and continuous-time analogues are given. Closely related results were proven by A. De Masi et al. (cf. their paper [Particle systems, random

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- Is it well structured, is it built up in a logical manner? We'll see parts of a paper next.

Parts of a good paper

This might well depend on the field, so here are my personal and probability-related views:

- Informative title. (If it can be short as well that's even better.)

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- Concise but descriptive abstract, Preferably *no formulas, no citations here*.

"Someone told me that each equation I included in the book would halve the sales."

Stephen Hawking
A Brief History of Time

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Except when not. . .

IOP FTC

IOP PUBLISHING JOURNAL OF PHYSICS A: MATHEMATICAL AND THEORETICAL

J. Phys. A: Math. Theor. **44** (2011) 492001 (5pp) doi:10.1088/1751-8113/44/49/492001

FAST TRACK COMMUNICATION

Can apparent superluminal neutrino speeds be explained as a quantum weak measurement?

M V Berry[1], **N Brunner**[1], **S Popescu**[1] **and P Shukla**[2]

[1] H H Wills Physics Laboratory, Tyndall Avenue, Bristol BS8 1TL, UK
[2] Department of Physics, Indian Institute of Technology, Kharagpur, India

Received 12 October 2011, in final form 27 October 2011
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Abstract
Probably not.

PACS numbers: 03.65.Ta, 03.65.Xp, 14.60.Pq

If recent measurements [1] suggesting that neutrinos travel faster than light survive scrutiny, the question of their theoretical interpretation will arise. Here we discuss the possibility that the apparent superluminality is a quantum interference effect, that can be interpreted as a weak measurement [2–5]. Although the available numbers strongly indicate that this explanation is not correct, we consider the idea worth exploring and reporting—also because it might suggest interesting experiments, for example on electron neutrinos, about which relatively little is known. Similar suggestions, though not interpreted as a weak measurement [6, 7] or not accompanied by numerical estimates [6, 8], have been proposed independently.

The idea, following analogous theory and experiment [9] involving light in a birefringent optical fibre, is based on the fact that the vacuum is birefringent for neutrinos. We consider the initial choice of neutrino flavour as a preselected polarization state, together with a spatially

Z. Wahrscheinlichkeitstheorie verw. Gebiete
55, 119–122 (1981)

Zeitschrift für
Wahrscheinlichkeitstheorie
und verwandte Gebiete
© Springer-Verlag 1981

An Elementary Proof of the Strong Law of Large Numbers

N. Etemadi

Mathematics Department, University of Illinois at Chicago Circle, Box 4348, Chicago IL 60680, USA

Summary. In the following note we present a proof for the strong law of large numbers which is not only elementary, in the sense that it does not use Kolmogorov's inequality, but it is also more applicable because we only require the random variables to be pairwise independent. An extension to separable Banach space-valued r-dimensional arrays of random vectors is also discussed. For the weak law of large numbers concerning pairwise independent random variables, which follows from our result, see Theorem 5.2.2 in Chung [1].

Theorem 1. *Let $\{X_n\}$ be a sequence of pairwise independent, identically distributed random variables. Let $S_n = \sum_{i=1}^{n} X_i$. Then*

$$E|X_1| < \infty \Rightarrow \lim_{n\to\infty} \frac{S_n}{n} = EX_1 \quad \text{a.s.}$$

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  - conclusion if there is one.

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Although it's still the authors' responsibility to make sure the arguments are rigorous and correct.

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Make notes. Preferably in the computer, because that stays. (Always have backup(s)!)

- Text-based (personal favourite: `Org mode...`)
- LaTeX note
- Paper + scan, or tablet (not searchable...)

And provide enough details: make sure *Future You* can make sense of your notes 5 years later.

Once done, think about the paper. Then ask yourself: have I achieved what I wanted from this paper? Are there parts worth revisiting for a second look? Deep understanding might require 2...3 readings.

Ok, so how to read it?

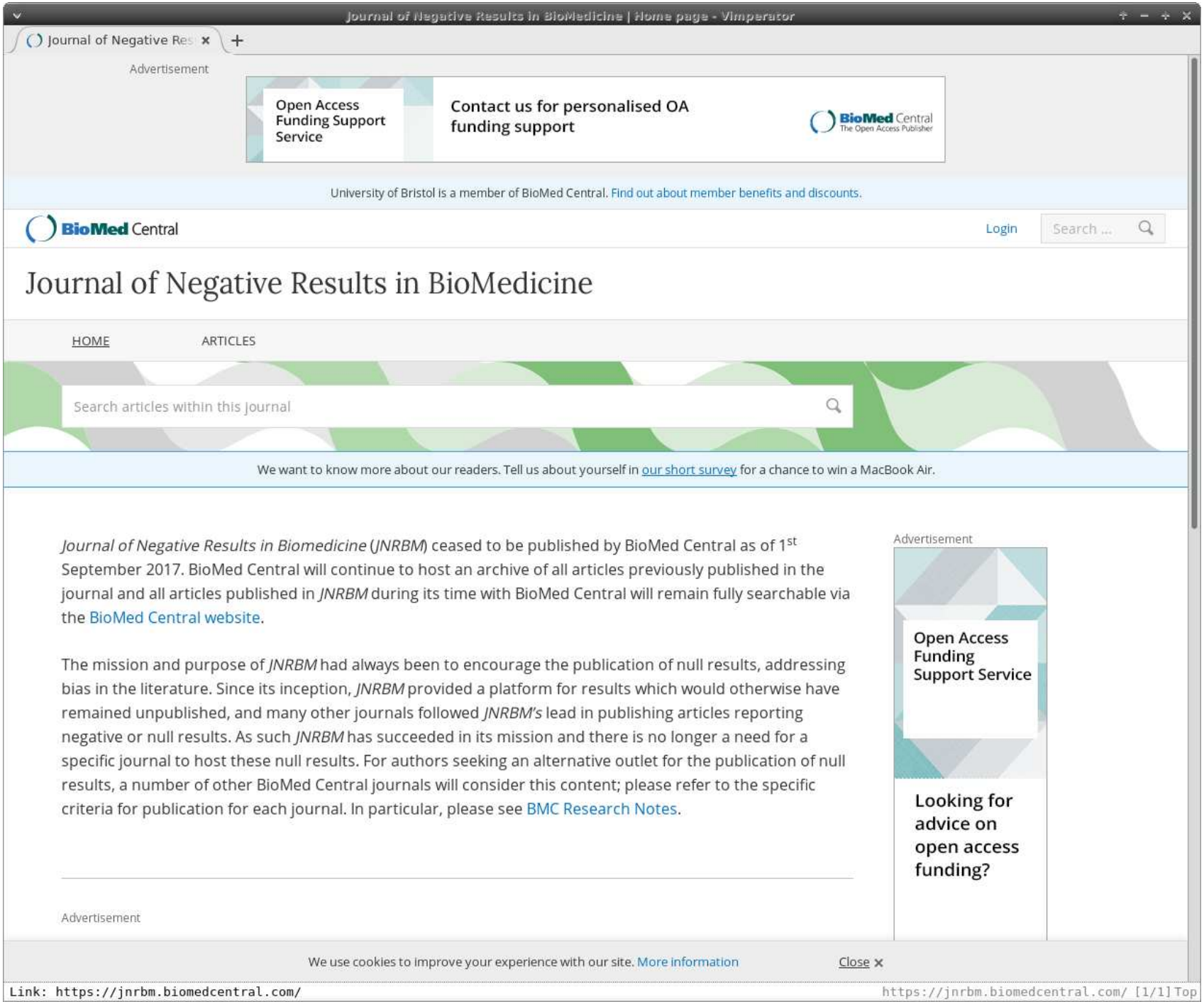
Still it might be difficult to read. Spend some time on it, maybe look at papers that cite this one or those this one cites. Then ask

- Advisor
- Fellow students
- Postdocs or other staff
- Author only as last resort. Be polite and very precise with your question.

Ok, so how to read it?

Many ideas will arise, usually more than you can immediately handle. Make a note of them (again, notes system?) and categorise. E.g.,

- Absolutely great idea, put in todo list to be looked at soon.
- This seems doable, will make a paper. . . eventually.
- This is an interesting but risky idea, should take a look at some point, maybe. . .
- This needs further reading and background check before I can decide on the next step.
- This has been done. :-)
- This has failed. Or others were faster. :-(Importance of negative results!


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Journal of Negative Results in Biomedicine (JNRBM) ceased to be published by BioMed Central as of 1st September 2017. BioMed Central will continue to host an archive of all articles previously published in the journal and all articles published in JNRBM during its time with BioMed Central will remain fully searchable via the BioMed Central website.
The mission and purpose of JNRBM had always been to encourage the publication of null results, addressing bias in the literature. Since its inception, JNRBM provided a platform for results which would otherwise have remained unpublished, and many other journals followed JNRBM's lead in publishing articles reporting negative or null results. As such JNRBM has succeeded in its mission and there is no longer a need for a specific journal to host these null results. For authors seeking an alternative outlet for the publication of null results, a number of other BioMed Central journals will consider this content; please refer to the specific criteria for publication for each journal. In particular, please see BMC Research Notes.
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JOURNAL OF NEGATIVE RESULTS

The primary intention of Journal of Negative Results is to provide an online-medium to publish peer-reviewed, sound scientific work in ecology and evolutionary biology that is scientifically rigorous but does not rely upon arbitrary significance thresholds to support conclusions. In recent years, the trend has been to publish only studies with 'significant' results and to ignore studies that seem uneventful. This may lead to a biased, perhaps untrue, representation of what exists in nature. By counter-balancing such selective reporting, JNR aims to expand the capacity for formulating generalizations. The work to be published in JNR will include studies that 1) test novel or established hypotheses/theories that yield negative or dissenting results, or 2) replicate work published previously (in either cognate or different systems). Short notes on studies in which the data are biologically interesting but lack statistical power are also welcome. JNR also intends to present the results of studies in a format suitable for formal meta-analysis. Research quality is of highest importance for JNR. Manuscripts will be assessed for publication on this basis - positive results or support for current scientific dogma are not essential.

'to do science is to search for repeated patterns' (MacArthur, 1972)

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VOL 11, NO 1 (2016)

TABLE OF CONTENTS

ARTICLES

Assemblage, size and abundance bias in a novel sandy shore macro-infaunal sampling technique — ABSTRACT PDF
Natalie Sheppard, Michael A. Weston, Sarah Butler, Bob Baird, Peter Dann

Defensive role of allelopathic secondary compounds in plants I: testing two independent general hypotheses — ABSTRACT PDF
Isabel P Pereira, Alexandra S Dias, Luís S Dias

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http://www.jnr-eeb.org/index.php/jnr [1/1] All

And how to write a good paper?

Need a good and relevant research idea that either solves an open problem that people are interested in, or one that people will be interested in. (This can take decades. . .)

Then: see *parts of a good paper*.

And how to write a good paper?

Ethics: “To the best of your knowledge”.

- Plagiarism is a very serious issue (many institutions use Turnitin).
- No intentional gaps or errors in proofs or research data.
- No making up of data.
- Avoid the word *obviously* :-) And make sure you sorted all details behind *easy to check that*.
- Acknowledge sources. Cite papers, thank discussion partners if not coauthors. Check copyright issues of figures and tables if not your original ones.
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